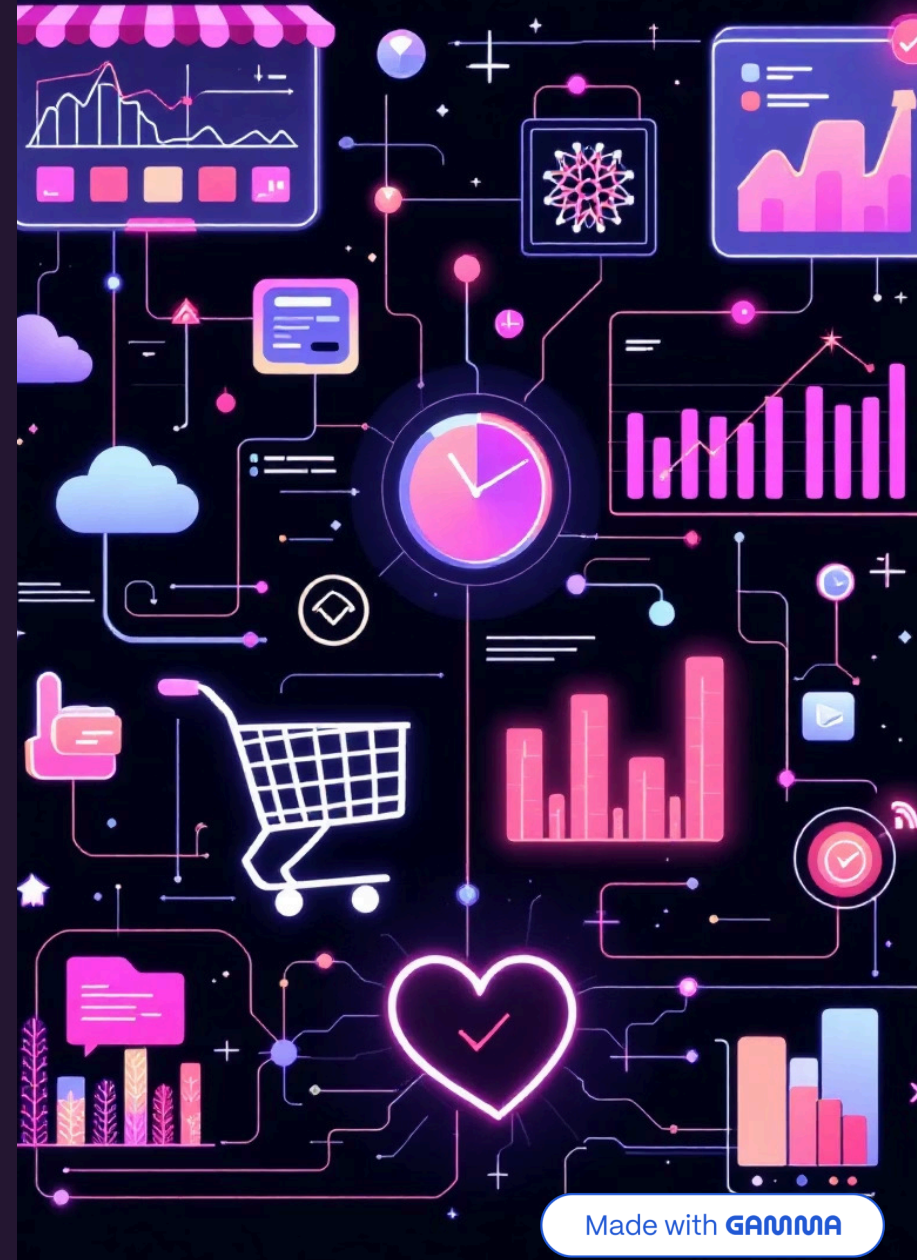
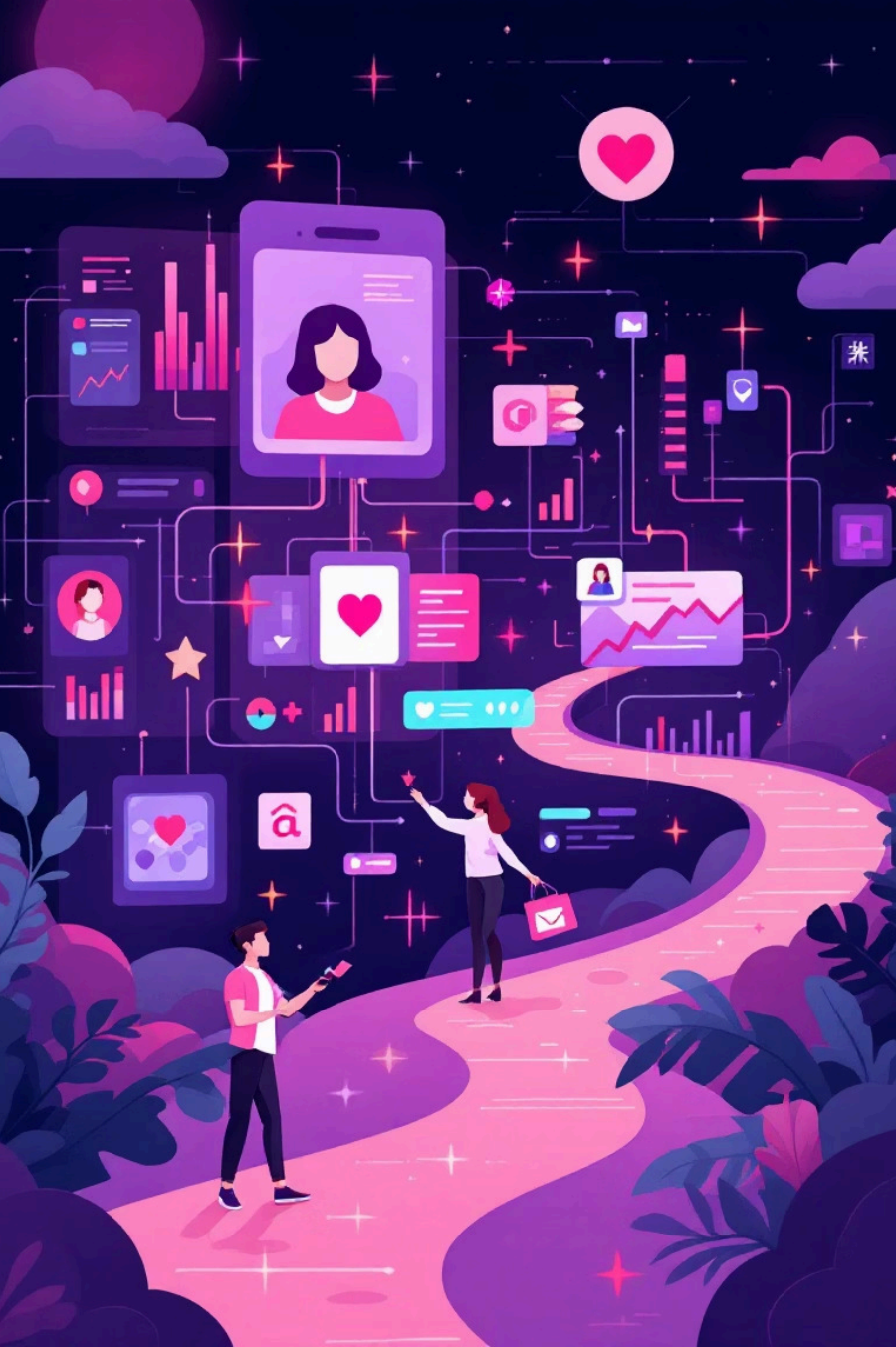


# Customer Shopping Behavior Analysis

Uncovering insights from transactional data to guide strategic business decisions.





# Project Overview & Data Summary

## Project Goal

Analyze 3,900 purchases to understand spending patterns, segments, and preferences.

## Dataset Snapshot

18 columns, including demographics, purchase details, and shopping behavior.

## Key Features

Age, Gender, Category, Purchase Amount, Review Rating, Subscription Status.

# Exploratory Data Analysis (EDA) in Python

## Data Preparation

- Loaded dataset with pandas.
- Initial exploration: `df.info()`, `df.describe()`.
- Missing data handled: Review Rating imputed by category median.
- Columns standardized to snake\_case.





# Feature Engineering & Consistency

12

## Age Group

Created binned age categories.



## Purchase Frequency

Mapped frequency to days.



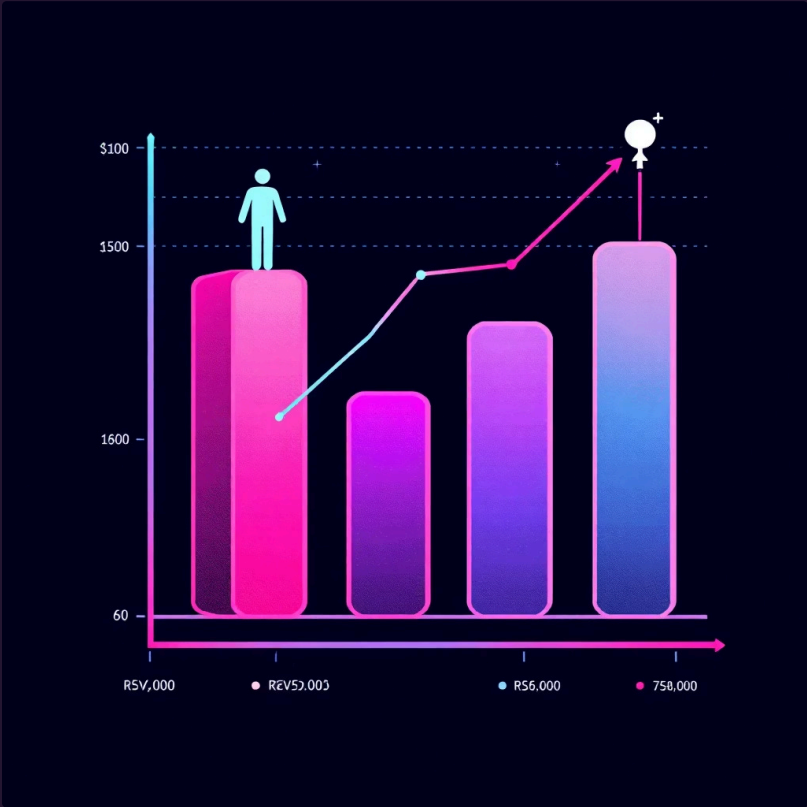
## Data Consistency

Dropped redundant 'promo\_code\_used' column.

# SQL Analysis: Revenue by Gender

Comparing total revenue generated by male vs. female customers.

|        |        |
|--------|--------|
| Male   | 157890 |
| Female | 75191  |





# SQL Analysis: High-Spending Discount Users & Top Products

1

## Discount Users

Identified customers using discounts but spending above average.

2

## Top 5 Products

Gloves, Sandals, Boots, Hat, Handbag have highest average review ratings.

# SQL Analysis: Shipping & Subscriptions

## Shipping Type Comparison

Average purchase amounts for Standard vs. Express shipping.

|          |       |
|----------|-------|
| Standard | 59.76 |
| Express  | 60.55 |

## Subscribers vs. Non-Subscribers

Non-subscribers have higher total revenue and average spend.

|     |       |        |
|-----|-------|--------|
| Yes | 59.49 | 62645  |
| No  | 59.87 | 170436 |



## Segmented customer profiles

Properly segmented customer profiles provide a clear understanding of the needs and preferences of different customer groups. This allows businesses to tailor their marketing and sales efforts to specific segments, resulting in more effective campaigns and higher conversion rates.



Discount rate



Customer profiles



Discounts

50%



50%

# SQL Analysis: Discount-Dependent & Customer Segmentation

## Discount-Dependent Products

Hat, Sneakers, Coat, Sweater, Pants have highest discount rates.

## Customer Segmentation

Loyal (3116), Returning (701), New (83) customers identified.



# SQL Analysis: Repeat Buyers & Revenue by Age

## Repeat Buyers & Subscriptions

Customers with >5 purchases are more likely to be non-subscribers.

|     |      |
|-----|------|
| Yes | 958  |
| No  | 2518 |

## Revenue by Age Group

Young Adults contribute the most revenue.

|             |       |
|-------------|-------|
| Young Adult | 62143 |
| Middle-aged | 59197 |
| Adult       | 55978 |
| Senior      | 55763 |

# Business Recommendations



## Boost Subscriptions

Promote exclusive benefits.



## Customer Loyalty

Reward repeat buyers.



## Review Discount Policy

Balance sales with margin.



## Targeted Marketing

Focus on high-revenue groups.

